

Golden Hour Grid

One map to prevent the next case and find the current one

Predictive Hotspots (Problem 12) + Missing Person Assistant (Problem 07) | IBM Bob AI Hackathon x NFSU

Team AlgoRhythms

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Problem statement

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Predictive hotspots

Patrols follow fixed routes even as crime patterns, events and trends shift week to week.

07

Missing person assistant

A case starts with scattered tips, CCTV notes and paperwork. Contradictory tips waste hours.

80,000+

children go missing in India every year

Nothing links the search to the risks in the area: lost hours, wasted patrol effort, no clear plan for the SHO. (NCRB 2022, per hackathon brief)

Gap: existing vs our solution

Existing approach

Fixed patrol routes

Manual sorting of tips and CCTV notes

Search area guessed by experience

Crime and missing-person work kept separate

Paperwork written by hand

Golden Hour Grid

Weekly top-5 micro-zones from recent trends, time of day and event calendar

Leads parsed, scored and checked for time-and-distance feasibility

Probability grid that grows with hours elapsed

One fused map: likely AND high-risk cells get priority

Brief, public appeal and case file drafted in one click

Our solution: Prevent, Search, Act



Prevent

Top 5 at-risk micro-zones for the coming week, each with reasons and event-linked spikes.
Backtested against a same-as-last-week baseline.



Search

Search area that grows with time and ranked leads. Contradictory or impossible tips are down-ranked before resources are committed.

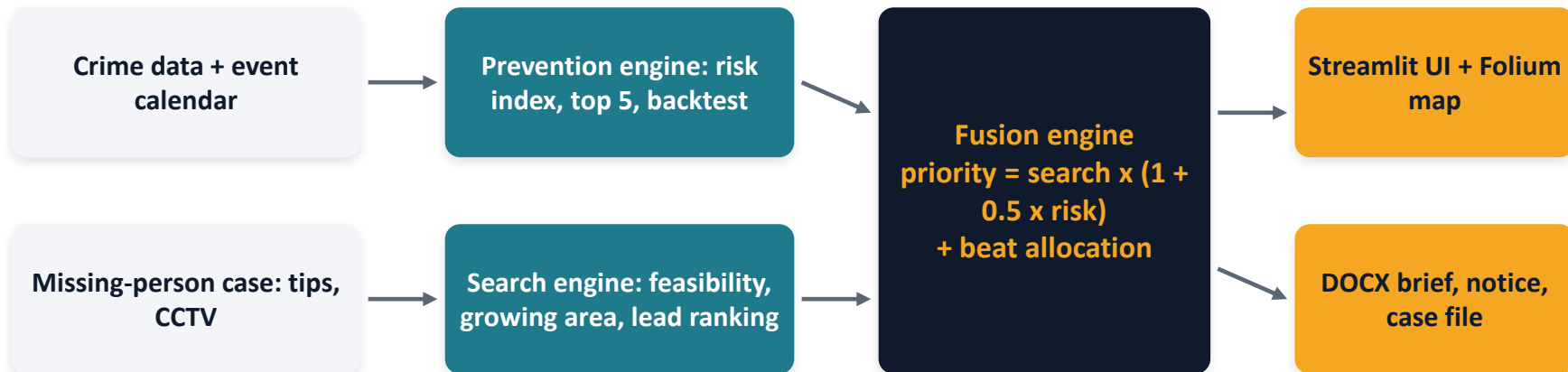


Act

Fused risk + search map, patrol beat allocation with time windows, plus SHO brief, public appeal and case file.

Synthetic data. Decision support for officers, not a replacement.

Architecture and working flow



AI layer (llm.py, template fallback): structures tips, explains rankings, drafts documents. Code computes every number, AI writes and explains.

Flow: 1 Ingest data > 2 Score risk and rank leads > 3 Fuse layers > 4 Allocate beats > 5 Generate documents

Future scope

1

Real data

Integrate CCTNS / ZIPNET records in place of synthetic data

2

Calibrated probabilities

Validate and calibrate risk and search probabilities on real cases

3

Hindi and regional tips

Accept and parse tips in Hindi and other local languages

4

Audit logs

Record every score, ranking and document for accountability

5

Live camera feeds

Link CCTV networks so sightings arrive automatically

6

Field app

Push beat assignments and updates to officers on mobile